2024 Annual Drinking Water Quality Report
Town of Keysville- PWSID# 5037300

INTRODUCTION
This Annual Drinking Water Quality Report for calendar year 2024 is designed to inform you about your drinking water quality. Our goal is to provide you with a safe and dependable supply of drinking water, and we want you to understand the efforts we make to protect your water supply. The quality of your drinking water must meet state and federal requirements administered by the Virginia Department of Health (VDH).

If you have questions about this report, or if you want additional information about any aspect of your drinking water or want to know how to participate in decisions that may affect the quality of your drinking water, please contact: Michael Funderburk, Director of Operations at 434-955-1045 or William Samples at 434-579-2564.

GENERAL INFORMATION
The sources of drinking water (both tap and bottled water) include rivers, lakes, streams, ponds, reservoirs, springs, and wells. As water travels over the surface of the land or through the ground, it dissolves naturally occurring minerals and, in some cases, radioactive material, and can pick up substances resulting from the presence of animals or from human activity. Contaminants that may be present in source water include:

- Microbial contaminants, such as viruses and bacteria, which may come from sewage treatment plants, septic systems, agricultural livestock operations, and wildlife.

- Inorganic contaminants, such as salts and metals, which can be naturally-occurring or result from urban stormwater runoff, industrial or domestic wastewater discharges, oil and gas production, mining or farming.

- Pesticides and herbicides, which may come from a variety of sources such as agriculture, urban stormwater runoff, and residential uses.

- Organic chemical contaminants, including synthetic and volatile organic chemicals, which are byproducts of industrial processes and petroleum production, and can also come from gas stations, urban stormwater runoff, and septic systems.

- Radioactive contaminants, which can be naturally occurring or be the result of oil and gas production and mining activities.

In order to ensure that tap water is safe to drink, EPA prescribes regulations that limit the amount of certain contaminants in water provided by public water systems. Food and Drug Administration regulations establish limits for contaminants in bottled water, which must provide the same protection for human heath.

All drinking water, including bottled drinking water, may reasonably be expected to contain at least small amounts of some contaminants. The presence of contaminants does not necessarily indicate that water poses a health risk. More information about contaminants and potential health effects can be obtained by calling the Environmental Protection Agency's Safe Drinking Water Hotline (800-426-4791).

Turbidity has no health effects. However, turbidity can interfere with disinfection and provide a medium for microbial growth. Turbidity may indicate the presence of disease-causing organisms. These organisms include bacteria, viruses, and parasites that can cause symptoms such as nausea, cramps, diarrhea and associated headaches.

Some people may be more vulnerable to contaminants in drinking water than the general population. Immuno-compromised persons such as persons with cancer undergoing chemotherapy, persons who have undergone organ transplants, people with HIV/AIDS or other immune system disorders, some elderly, and infants can be particularly at risk from infections. These people should seek advice about drinking water from their health care providers. EPA/CDC guidelines on appropriate means to lessen the risk of infection by cryptosporidium and other microbiological contaminants are available from the Safe Drinking Water Hotline (800-426-4791).

Lead can cause serious health problems, especially for pregnant women and young children. Lead in drinking water is primarily from materials and components associated with service lines and home plumbing. The Town of Keysville is responsible for providing high quality drinking water and removing lead pipes, but cannot control the variety of materials used in plumbing components in your home. You share the responsibility for protecting yourself and your family from the lead in your home plumbing. You can take responsibility by identifying and removing lead materials within your home plumbing and taking steps to reduce your family's risk. Before drinking tap water, flush your pipes for several minutes by running your tap, taking a shower, doing laundry or a load of dishes. You can also use a filter certified by an American National Standards Institute accredited certifier to reduce lead in drinking water. If you are concerned about lead in your water and wish to have your water tested, contact The Town of Keysville and 434-736-9551. Information on lead in drinking water, testing methods, and steps you can take to minimize exposure is available at http://www.epa.gov/safewater/lead

SOURCE OF YOUR DRINKING WATER

The source of your drinking water is Spring Creek Impoundment located in Keysville Virginia.
Treatment of the raw water consists of chemical addition, coagulation, flocculation, settling, filtration, fluoridation and chlorination. All of these processes work together to remove the physical, chemical, and biological contaminants to make the water safe for drinking.

A source water assessment of our system was conducted in 2002 by the Virginia Department of Health. The assessment report consists of maps showing the source water assessment area, an inventory of known land use activities of concern, and documentation of any known contamination within the last 5 years. The report is available by contacting your water system representative at the phone number or address given elsewhere in this drinking water quality report.

DEFINITIONS

Contaminants in your drinking water are routinely monitored according to Federal and State regulations. The table on the next page shows the results of our monitoring for calendar year 2023. In the table and elsewhere in this report you will find many terms and abbreviations you might not be familiar with. The following definitions are provided to help you better understand these terms:

Non-detects (ND) - lab analysis indicates that the contaminant is not present within the detection limits of the instrument used.

Parts per billion (ppb) or Micrograms per liter (ug/l) - one part per billion corresponds to one minute in 2,000 years, or one penny in $10,000,000.

Parts per million (ppm) or Milligrams per liter (mgl) - one part per million corresponds to one minute in two years or one penny in $ 10,000.

Parts per trillion (ppt) or Nanograms per liter (ngl) - one part per trillion corresponds to one minute in 2,000,000 years or one penny in $10,000,000,000.

Picocuries per liter (pCi/L) - picocuries per liter is a measure of the radioactivity in water.

Nephelometric Turbidity Unit (NTU) - nephelometric turbidity unit is a measure of the cloudiness of the water. Turbidity in excess of 5 NTU is just noticeable to the average person.

Action Level (AL) - the concentration of a contaminant, which, if exceeded, triggers treatment or other requirements, which a water system must follow.

Treatment Technique (TT) - a required process intended to reduce the level of a contaminant in drinking water.

Maximum Contaminant Level Goal, or MCLG - the level of a contaminant in drinking water below which there is no known or expected risk to health. MCLGs allow for a margin of safety.

Maximum Contaminant Level, or MCL - the highest level of a contaminant that is allowed in drinking water. MCLs are set as close to the MCLGs as feasible using the best available treatment technology.

Maximum residual disinfection level goal (MRDLG) – the level of a drinking water disinfectant below which there is no known or expected risk to health. MRDLGs do not reflect the benefits of the use of disinfectants to control microbial contaminants.

Maximum residual disinfectant level (MRDL) – the highest level of a disinfectant allowed in drinking water. There is convincing evidence that addition of a disinfectant is necessary for control of microbial contaminants.

Abbreviations: NA – Not Applicable.

WATER QUALITY RESULTS

Inorganic Contaminants						
Contaminant / Unit of Measurement	**MCLG**	**MCL**	**Level Found / Range**	**Exceedance**	**Date of Sample**	**Typical Source of Contamination**
Nitrates ppm	10	10	0.06	No	July 2024	Runoff from fertilizer use; Leaching from septic tanks, sewage; Erosion of natural deposits.
Lead ppb	0	AL = 15	8.1 (90th percentile) Range: ND – 11.6 Of the 10 samples collected 0 exceeded the AL.	No	September 2023	Corrosion of household plumbing systems; Erosion of natural deposits
Copper ppm	1.3	AL=1.3	0.0684 (90th percentile) Range: ND to 0.0968 Of the 10 samples collected 0 exceeded the AL.	No	September 2023	Corrosion of household plumbing systems; Erosion of natural deposits
Fluoride ppm	4	4	Average: 0.61 Range: 0.26-0.95	No	Monthly 2024	Erosion of natural deposits; water additive which promotes strong teeth; discharge from fertilizer and aluminum factories
Microbiological Contaminants						
Contaminant / Unit of Measurement	**MCLG**	**MCL**	**Level Found / Range**	**Exceedance**	**Date of Sample**	**Typical Source of Contamination**
Turbidity / NTU	N/A	1.0 Max TT 0.3 in 95% of monthly samples	0.07 NTU MAX CFE 100% <0.3	No	Daily 2024	Soil Runoff

Total Coliform Bacteria	NA	TT	0	No	2024	the environment

Disinfection Byproducts						
Contaminant / Unit of Measurement	**MCLG**	**MCL**	**Level Found / Range**	**Exceedance**	**Date of Sample**	**Typical Source of Contamination**
Chlorine ppm	MRDLG = 4	MRDL = 4.0	Avg. 1.8 Range: 1.5 – 3.1	No	Daily Entry Point	Water additive used to control microbes
HAA5s (Total Haloacetic Acids) ppb	N/A	60	Highest Qtr. 43 Range: 22-77	No	Quarterly 2024	By-product of drinking water disinfection
TTHMs (Total Trihalomethanes) ppb	N/A	80	Highest Qtr. 87 Range: 58-111	Yes*	Quarterly 2024	By-product of drinking water disinfection
Total Organic Carbon (TOC) ppm	N/A	TT-TOC removal ratio greater than or equal to 1.00	Lowest Quarterly Ave. Ratio: 1.30 Range: 1.20-1.58	No	Tested quarterly at raw and treated water	Naturally present in the environment
Non-Regulated Contaminants						
Contaminant / Unit of Measurement	**MCLG**	**MCL**	**Level Found / Range**	**Exceedance**	**Date of Sample**	**Typical Source of Contamination**
Perfluorobutanoic acid (PFBA) ppt	N/A	N/A	4.4	No	October 2023	Breakdown of other PFAS that have been used in stain-resistant fabrics, paper food packaging, carpets, and consumer products

Water Quality Results

We regularly monitor various contaminants in the water supply to meet all regulatory requirements. The table lists only those contaminants that had some level of detection. Many other contaminants have been analyzed but were not present or were below the detection limits of the lab equipment.

Most of the results in the table are from testing done in 2024. However, the state allows us to monitor for some contaminants less than once per year because the concentrations of these contaminants do not change frequently.

The sodium concentration in the treated water was detected at 24.8 ppm in July 2024. EPA's maximum recommended level of sodium is 20 ppm for people on a sodium-restricted diet.

MCL's are set at very stringent levels. To understand the possible health effects described for many regulated constituents, a person would have to drink 2 liters of water every day at the MCL level for a lifetime to have a one-in-a-million chance of having the described health effect.

A lead service line inventory has been completed in 2024 at which no lead lines were found. Customers can access a copy of the inventory by contacting the Town of Keysville at 434-736-9551.

Additional Required Health Effects Language

Some people who drink water containing trihalomethanes in excess of the MCL over many years may experience problems with their liver, kidneys, or central nervous systems, and may have an increased risk of getting cancer.

VIOLATION INFORMATION

The Town of Keysville exceeded the MCL for TTHMs in the third and fourth quarters of 2024. The public notices were mailed out in September and December of 2024 for the exceedances.

This Drinking Water Quality Report was prepared by Michael C. Funderburk Jr, Director of Operations, Samples' Monitoring Service, LLC at 1099 Bold Springs Road, South Boston Virginia 24592.

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